

SIMATS ENGINEERING

ASSESSMENT TOOL 3 — ANNEXURE A

ARTICLE REVIEW

Course: Artificial Intelligence (CSA17)

Course Outcome: CO4 — Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making.

Topics addressed: Decision Trees; Statistical Learning Methods (Logistic Regression)

Task 1: Article Summary

(a) Full Citation

Joshi, R. D., & Dhakal, C. K. (2021). Predicting Type 2 Diabetes Using Logistic Regression and Machine Learning Approaches. *International Journal of Environmental Research and Public Health*, 18(14), 7346. <https://doi.org/10.3390/ijerph18147346>

Domain / Application Area: Healthcare analytics and clinical decision support — early screening for Type 2 Diabetes Mellitus (T2DM).

Specific ML Problem: A supervised binary classification problem — predicting whether a Pima Indian woman is diabetic or non-diabetic from clinical and demographic attributes, and identifying which attributes drive that prediction.

(b) Summary of Objective and Research Gap (own words)

Diabetes is a widespread chronic disease that drives substantial morbidity, mortality, and healthcare cost, so early and accurate risk identification matters. The authors note that while many machine learning models chase raw predictive accuracy, this focus can come at the expense of clinical interpretability, leaving practitioners unsure which risk factors are actually driving a prediction. That tension between opaque, accuracy-first models and the clinician's need for transparent, actionable risk factors is the gap this paper responds to. Their approach is to fit a logistic regression model on the Pima Indian Diabetes dataset to statistically quantify each factor's contribution, and independently grow a classification (decision) tree on the same data purely to cross-check which variables the data naturally splits on first. Agreement between the two methods becomes the authors' evidence that the identified predictors are robust rather than an artifact of one modeling choice. Their analysis converges on five recurring predictors — glucose level, pregnancy count, BMI, diabetes pedigree function, and age — and their preferred logistic specification reaches about 78% classification accuracy, which they position as strong enough to underpin low-cost, non-invasive screening efforts.

Task 2: Methodology Analysis

(a) ML Technique(s) and Dataset

The paper's primary technique is logistic regression, a statistical classification model that estimates the log-odds of a binary outcome (diabetic vs. non-diabetic) as a linear combination of predictors, allowing each coefficient to be read as that variable's independent contribution to risk. Alongside this, the authors independently grow a classification decision tree (a CART-style recursive-partitioning model) on the same records, not to outperform the regression but to visually and non-parametrically confirm which variables the data itself considers most discriminating. They report two tree variants of different complexity: a compact six-

node tree that splits mainly on glucose, BMI, and age, and a larger ten-node tree that additionally brings in pregnancy count and diabetes pedigree function.

The dataset is the well-known Pima Indian Diabetes Dataset, which records clinical and demographic attributes — such as plasma glucose concentration, blood pressure, BMI, diabetes pedigree function, number of pregnancies, and age — for adult women of Pima Indian heritage, together with a binary label indicating a diabetes diagnosis. The authors validate their preferred logistic specification with cross-validation to guard against overfitting before reporting a single headline accuracy figure.

(b) Mapping to CO4 Topics, Strength, and Limitation

CO4 Topic	How the Article Maps to It
Statistical Learning Methods	Logistic regression is used as the primary, interpretable probabilistic classifier and the main basis for the paper's conclusions.
Decision Trees / Inductive Learning	A classification tree is inductively grown from the same records purely to validate, by an independent method, which attributes the regression identified as important.

Methodological Strength: Triangulating two structurally different techniques (a parametric statistical model and a non-parametric tree) on the same data is a genuine strength — agreement between them is stronger evidence for the five identified risk factors than either method would provide alone.

Methodological Limitation: The decision tree is used only descriptively, not tuned, pruned, or evaluated on its own held-out accuracy, so its role is closer to an interpretive aid than a rigorously benchmarked classifier — and the study never tests whether a properly optimized tree or ensemble might outperform the logistic model on this data.

Task 3: Results & Critical Evaluation

(a) Key Results

The authors report a single headline figure for their preferred logistic regression specification: a prediction accuracy of 78.26%, with a cross-validation error rate of 21.74%. The decision tree component is reported qualitatively rather than with its own accuracy score — its value lies in which variables it splits on, not in a competing performance number.

Model	Key Predictors Identified	Reported Performance
Logistic Regression (preferred model)	Glucose, pregnancy count, BMI, diabetes pedigree function, age	Accuracy: 78.26% CV error rate: 21.74%
Decision Tree (6-node)	Glucose, BMI, age	Not separately reported
Decision Tree (10-node)	Glucose, BMI, pregnancy, diabetes pedigree function, age	Not separately reported

(b) Critical Evaluation of the Results

The reported 78.26% accuracy is plausible for logistic regression on this dataset — comparable published work on the same Pima Indian Diabetes Dataset typically clusters accuracy in the high-70s to low-80s range

for linear models — so the figure is not surprising, but it is also not exceptional or clearly state-of-the-art. Two dataset-level biases are worth flagging: first, the Pima Indian Diabetes Dataset is restricted to adult women of one specific ethnic and genetic background, so any risk factor coefficients (and the accuracy figure itself) may not transfer to men or to other populations. Second, the dataset's class distribution is known to be imbalanced (roughly two-thirds non-diabetic to one-third diabetic in the standard version), which means accuracy alone can be a misleading headline metric — a classifier that is biased toward predicting the majority class can still post a fairly high accuracy while missing many true diabetic cases. The paper does not report precision, recall, sensitivity/specificity, F1-score, or AUC-ROC, all of which would show whether the model is missing diabetic cases disproportionately. Strengthening the study would mean reporting a full confusion matrix and AUC-ROC, testing on an external, more demographically diverse cohort, and comparing against a resampling technique (e.g., SMOTE) to check whether the class imbalance is depressing sensitivity.

(c) Real-World Applicability

A model built on easily obtained attributes (glucose reading, BMI, age, pregnancy history) is attractive for low-cost, non-invasive pre-screening in primary care or community health outreach, especially in settings without ready access to advanced laboratory testing. Logistic regression and small decision trees are also computationally trivial to deploy, requiring none of the infrastructure that deep learning would demand. The main challenges to real deployment are: data availability (a single-population, moderately sized historical dataset is a weak foundation for a clinical tool meant for broader populations), the ethical stakes of false negatives and false positives in a health-screening context, and the need for prospective clinical validation before any such model informs real diagnostic or treatment decisions.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights

- Decision trees can serve a validating, interpretive role alongside a statistical model, not only as standalone classifiers — this deepened my understanding of inductive learning as a way to inductively confirm feature importance (via recursive splits) independent of a model's parametric assumptions, directly connecting to the CO4 topic of Decision Trees / Inductive Learning.
- There is a real trade-off between interpretability and predictive power in Statistical Learning Methods: logistic regression is chosen here specifically because its coefficients are directly interpretable as risk-factor weights, even though more flexible techniques (SVM, ensembles) might classify more accurately — this connects to the CO4 topic of Statistical Learning Methods (Logistic Regression / SVM).
- A single accuracy figure is not sufficient evidence of a good classifier, especially on an imbalanced dataset — this reinforced why the course's classification and decision-making topics emphasize reporting precision, recall, F1-score, and AUC-ROC alongside accuracy, since the model's practical usefulness for decision-making depends on how errors are distributed between the classes, not just their total count.

(b) Proposed Extension

Proposed Experiment: Re-run the same risk-factor analysis using a gradient-boosted decision tree ensemble (e.g., XGBoost or Random Forest) combined with a class-balancing technique such as SMOTE, applied to a larger and more demographically diverse dataset such as NHANES (which includes men and multiple ethnic groups), while reporting a full metric suite (accuracy, precision, recall, F1-score, AUC-ROC) and validating on a held-out external cohort.

Feasibility: This is highly feasible — NHANES is a public, well-documented dataset, and tree-ensemble libraries (e.g., scikit-learn, XGBoost) are lightweight, well-supported, and require no specialized hardware, making this an incremental rather than a resource-intensive extension.

Expected Impact: A tree ensemble can capture non-linear interactions between risk factors that a linear logistic model cannot, class balancing should improve sensitivity to true diabetic cases, and testing across a broader population would directly address the generalizability limitation identified in Task 3(b) — together producing a more clinically trustworthy and reusable screening model.